



Horizontal Gene Transfer and Mobile Genetic Elements

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Learning Objectives

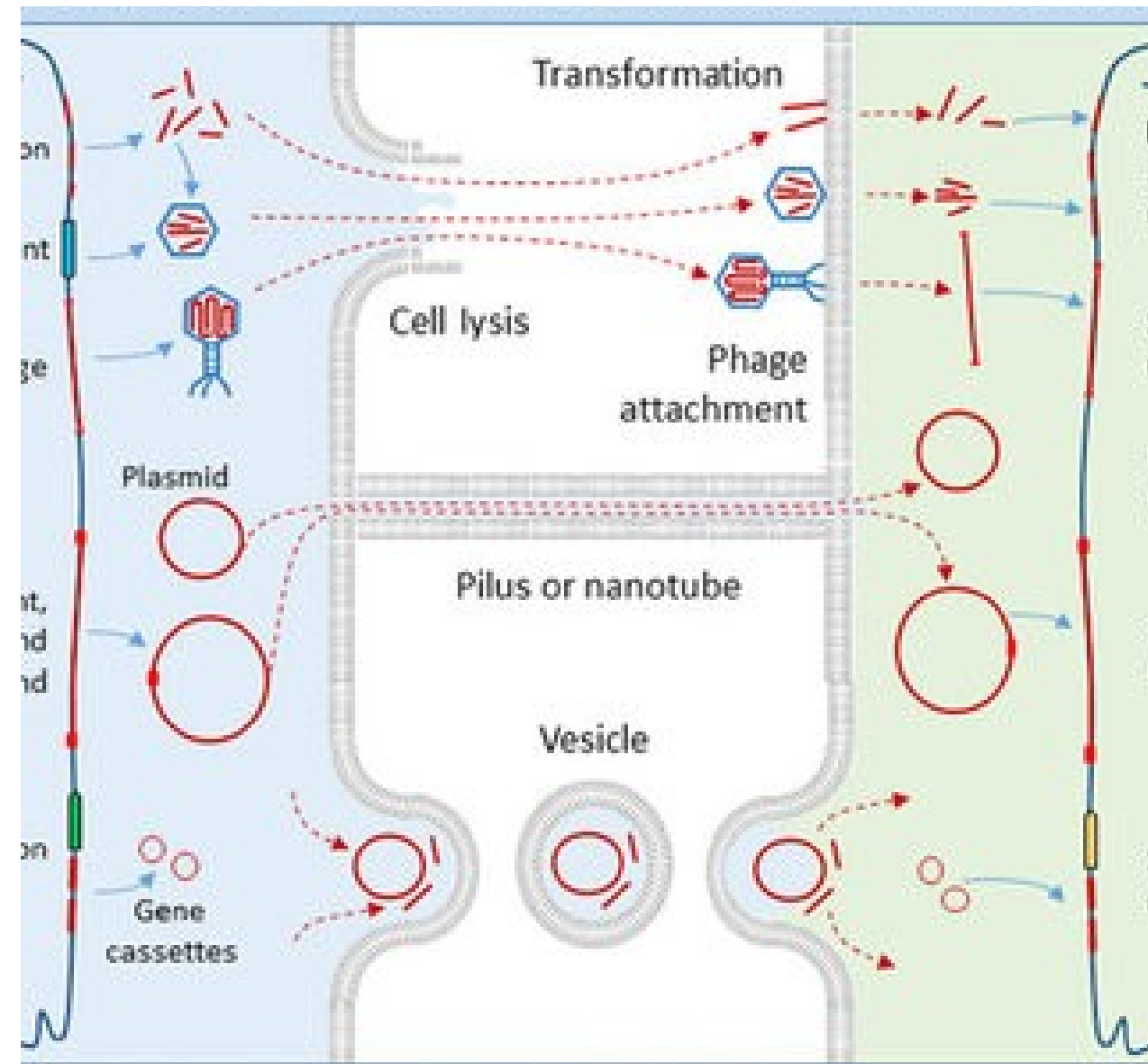
By the end of this module, participants will be able to:

- ✓ Understand the mechanisms of horizontal gene transfer
- ✓ Understand the diversity of mobile genetic elements (MGEs)
- ✓ Explain their roles in bacterial evolution and antimicrobial resistance (AMR)
- ✓ Distinguish between plasmids, IS elements, phages, and integrons
- ✓ Appreciate the interplay of MGEs in shaping bacterial genomes



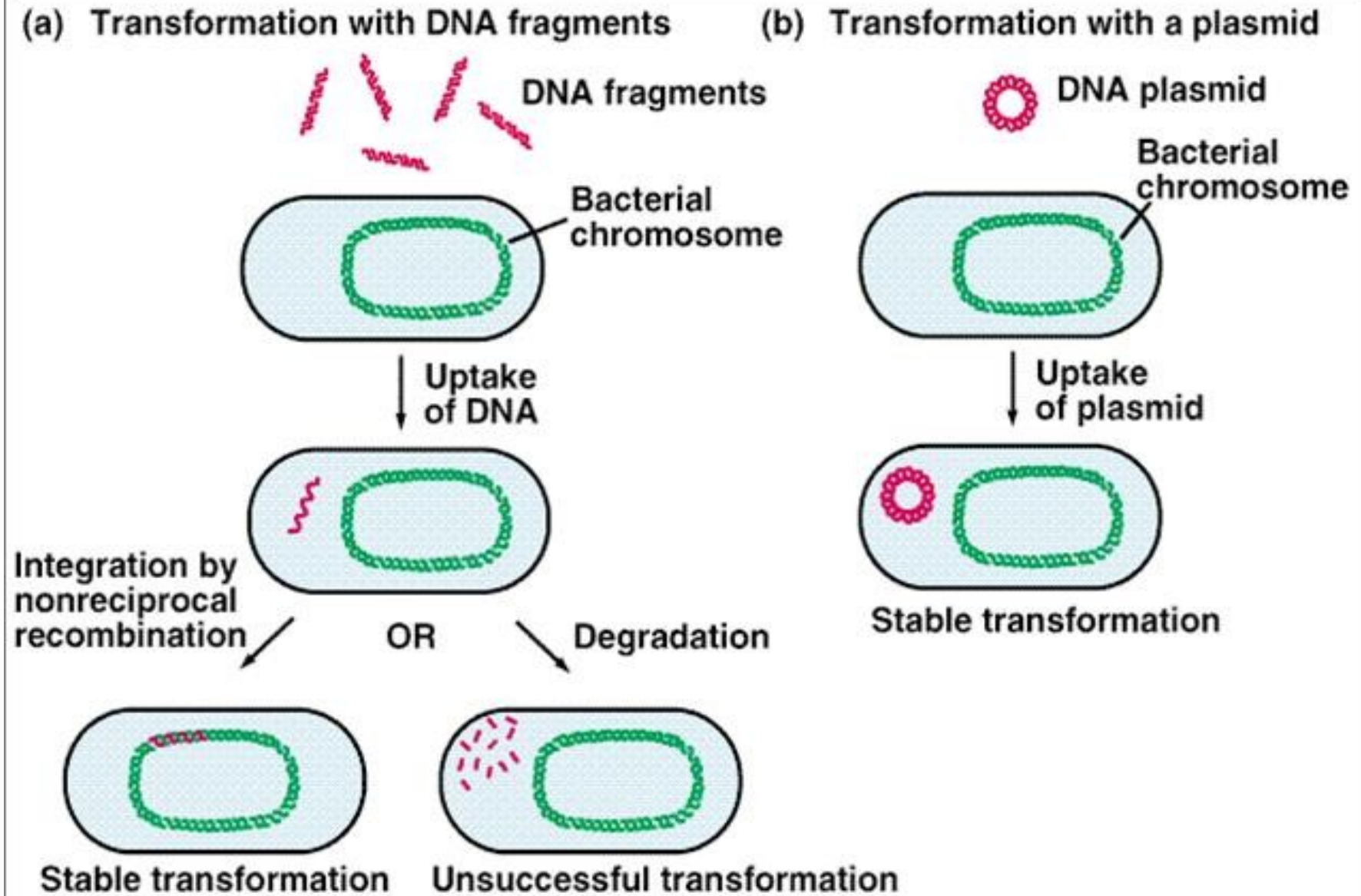
Horizontal Gene Transfer (HGT)

- **Horizontal vs. Vertical Inheritance:**
 - **Vertical inheritance**, where genes pass from parents to offspring,
 - **HGT** involves the transfer of genetic information across species boundaries.
- **Mechanisms:** Common methods of HGT include:
 - **Transformation:** The uptake of "naked" DNA from the environment.
 - **Transduction:** The transfer of genetic material by a virus (bacteriophage).
 - **Conjugation:** The direct transfer of genetic material between cells, often through plasmids, via a pilus or direct connection.



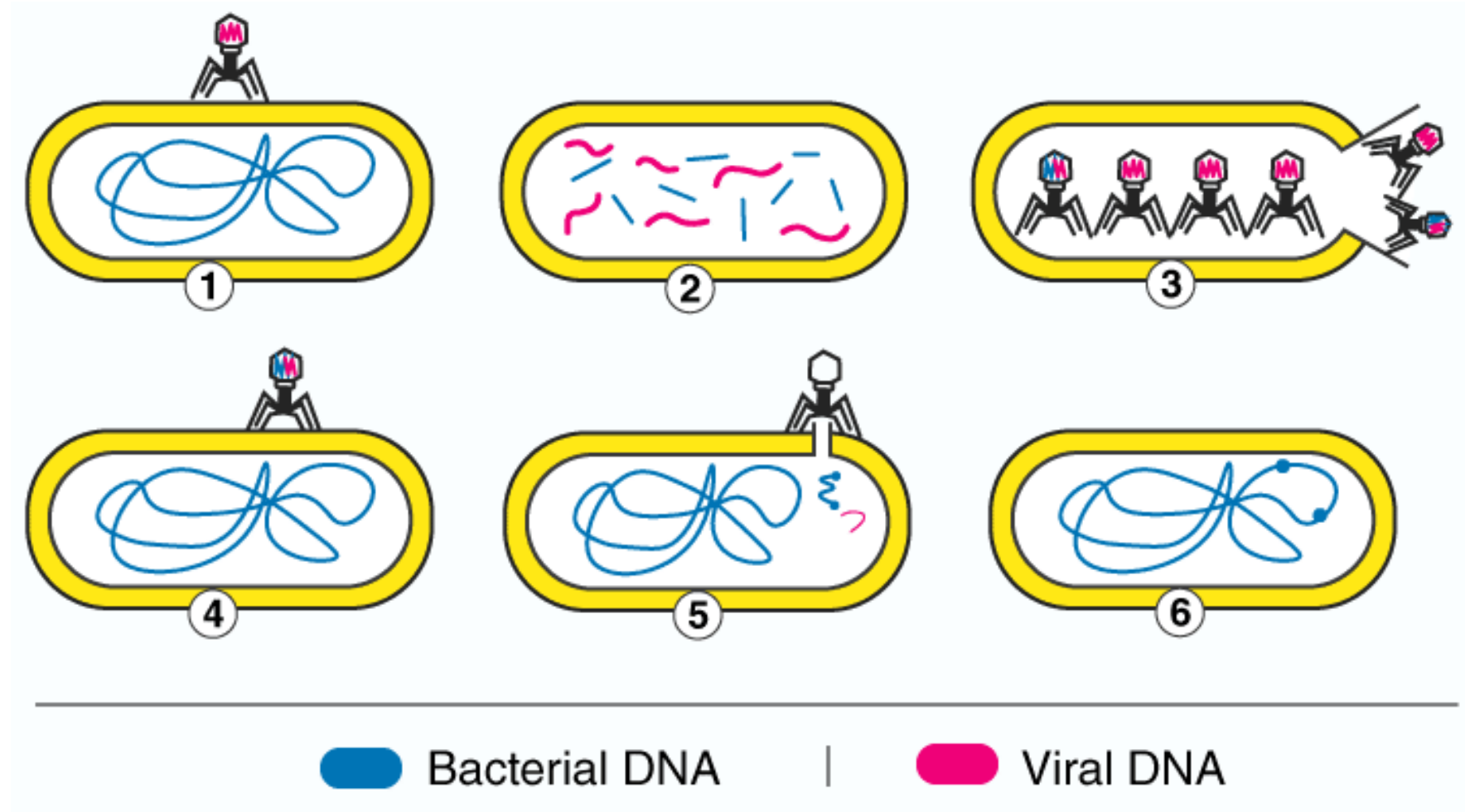
Horizontal Gene Transfer: Transformation

- Uptake of free DNA fragments from environment.
- Requires competence (e.g., *Neisseria* spp., *S. pneumoniae*, *H. pylori*).
- Example: penicillin resistance in *S. pneumoniae*.



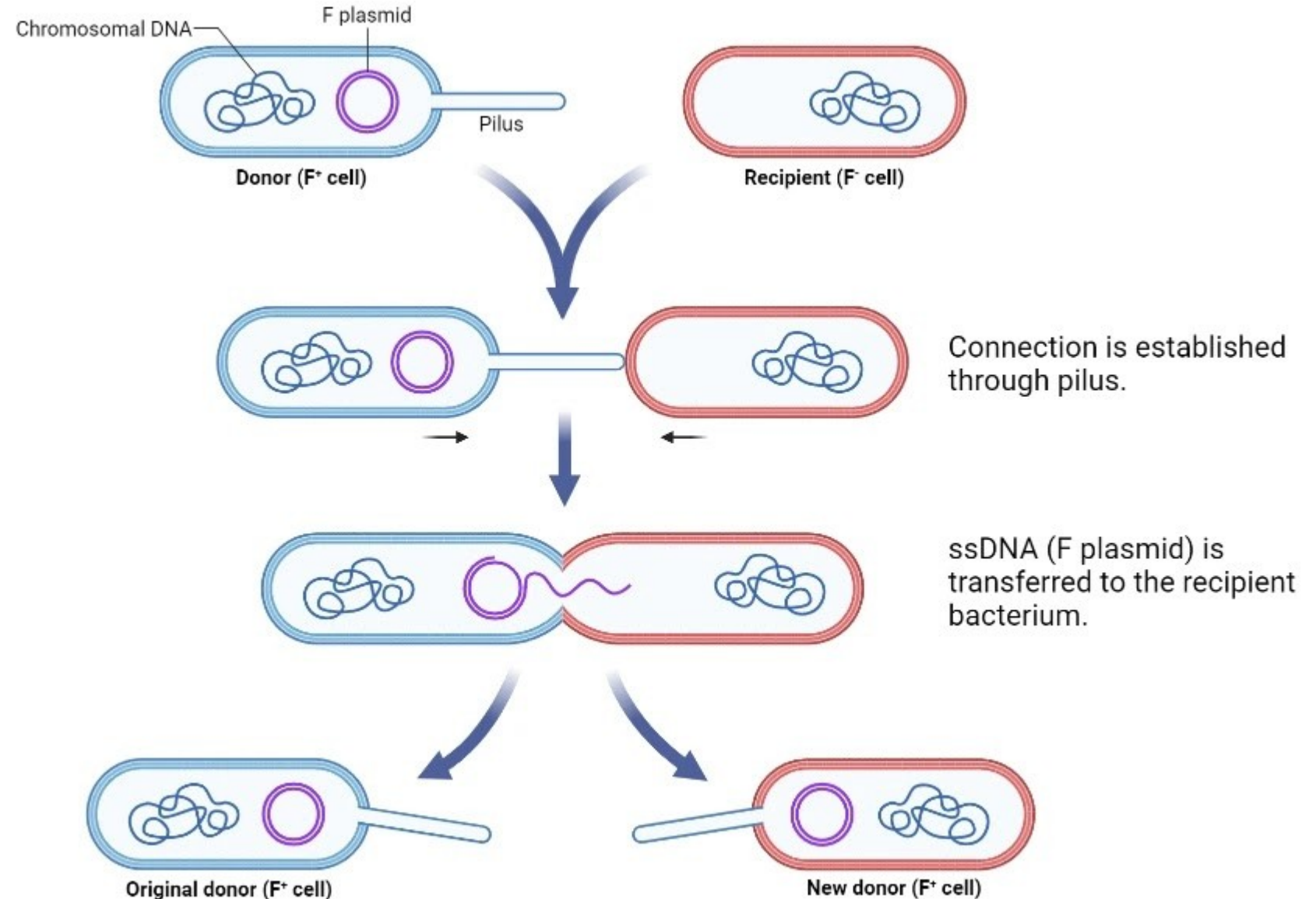
Horizontal Gene Transfer: Transduction

- DNA transfer via bacteriophages.
- Generalized: random bacterial DNA packaged.
- Specialized: specific genes transferred.
- Example: resistance + toxin genes in *S. aureus*.



Horizontal Gene Transfer: Conjugation

- Plasmid transfer via pilus.
- One strand transferred, replicated in recipient.
- Examples: ESBLs, carbapenemases (NDM, KPC)



Mobile Genetic Elements

- **Definition:** DNA elements that move within and between genomes
- **Importance:** Genome plasticity, horizontal gene transfer, adaptation
- **Key functions:** Resistance, virulence, metabolic capabilities

- These are sometimes referred to as **selfish genetic elements**.
- They are a type of genetic material that move around within the genome.

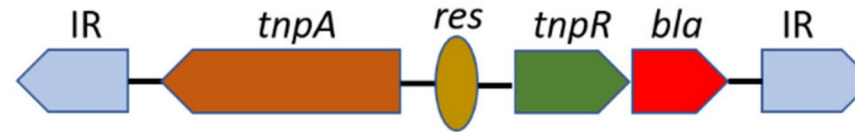
Examples include:

- Plasmids
- Transposons
- Integrons
- Bacteriophages
- Integrative and Conjugative Elements
- Insertion sequences.

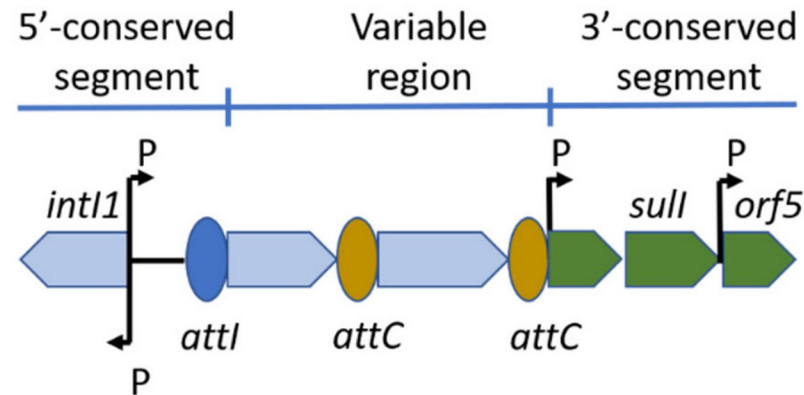
Mobile Genetic Elements

MGES

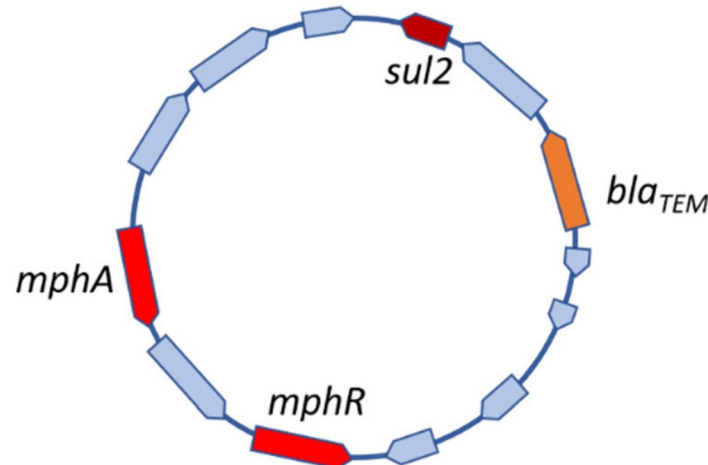
transposons



integrons

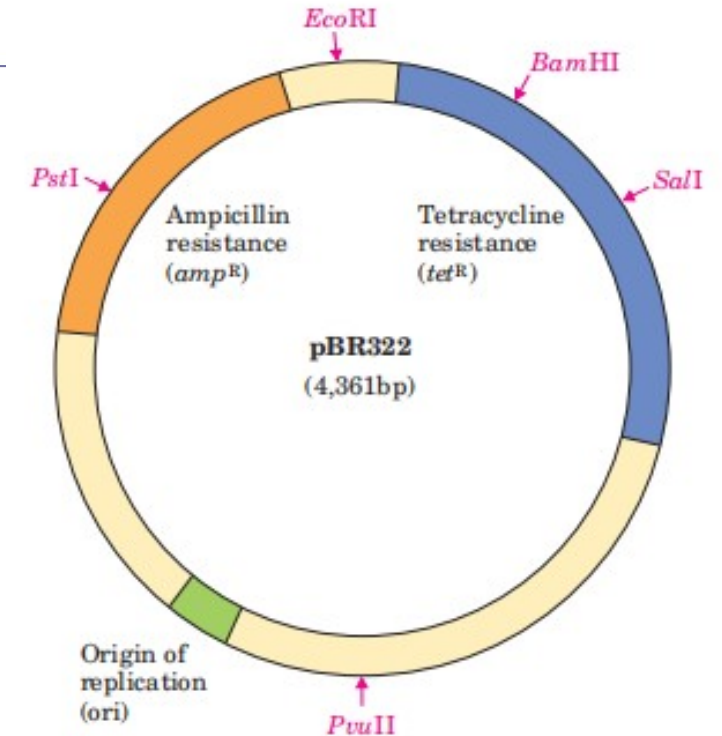


plasmids

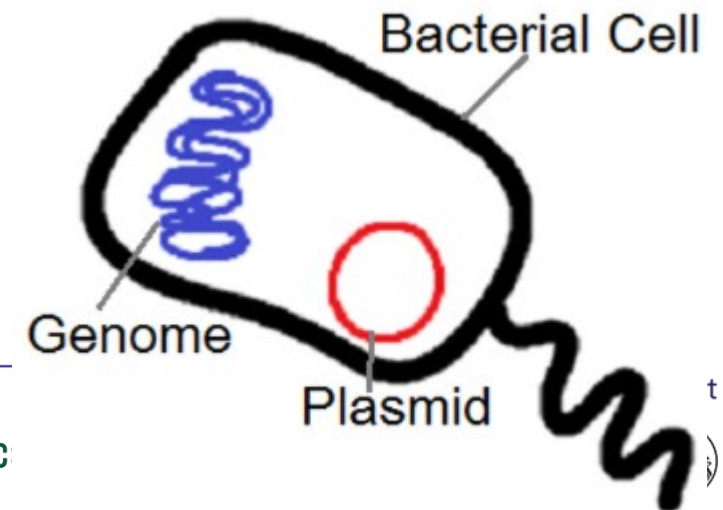


Plasmids

- Circular extra-chromosomal elements inside the bacterial cell.
- Not essential for survival but confer extra advantages to the cell.
- Some cells have none, some have many (20-30), others have multiple copies of the same plasmid.
- They replicate and multiply independently of the chromosome.
- They can be introduced into the cell via transformation.



The constructed E. coli plasmid pBR322



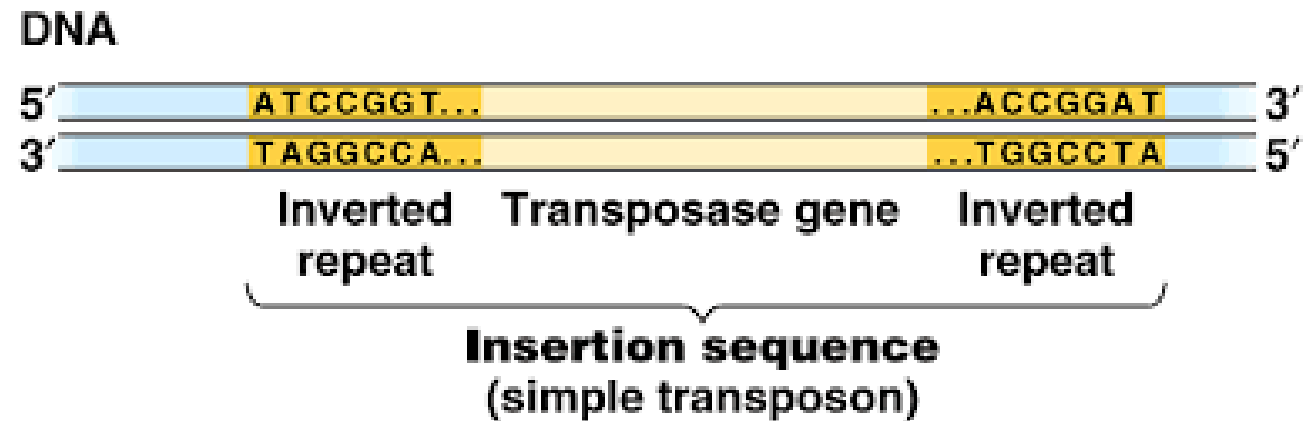
Plasmids

- Plasmid Transfer
 - **Conjugation:** pilus-mediated DNA transfer
 - **Mobilization:** with helper plasmids
 - **Transformation:** uptake of free plasmid DNA

- Case Study – Plasmids & AMR
 - Carbapenemase-producing Enterobacteriaceae (CPE)
 - Plasmid-encoded blaNDM-1, blaKPC genes
 - Global spread via conjugative plasmids

Insertion Sequences (IS Elements)

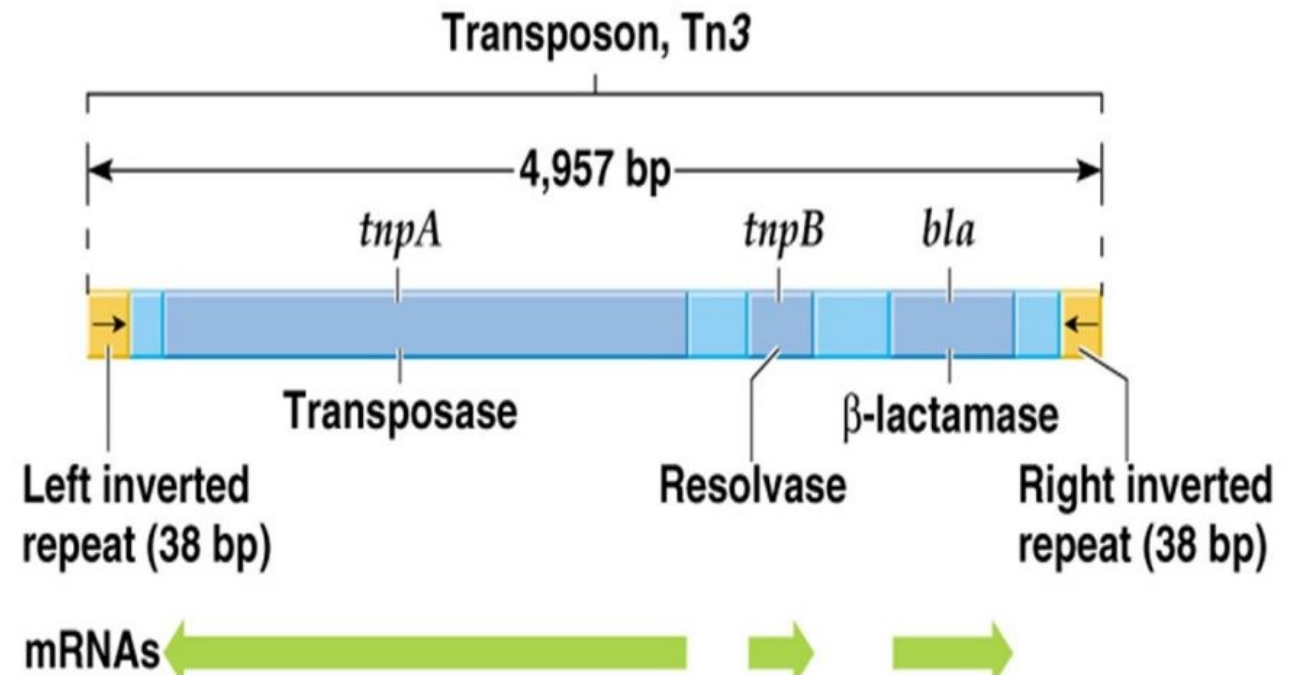
- A simple type of transposon consisting of only the sequences required for transposition.
- Cause gene disruption, activation, rearrangement
- Simplest MGEs (transposase + inverted repeats)
- Mechanisms: cut-and-paste, replicative transposition
- An example is the IS1 element in E.coli
- End of all known IS elements shows inverted Terminal Repeats (ITRs)



Transposons

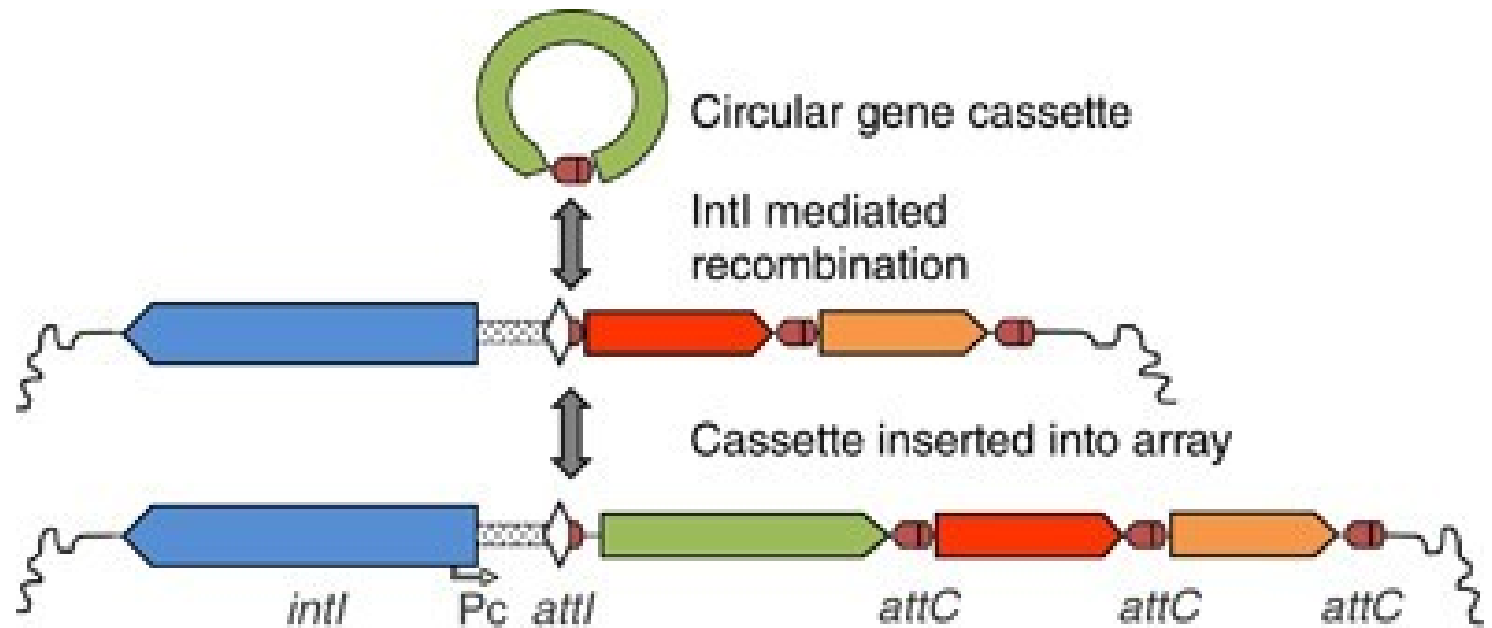
- DNA sequences that move from one location in the genome to another by either cutting and pasting themselves or by replicating and inserting copies elsewhere.
- Are a kind of molecular parasite, efficiently making a home within the host genome. Many have genes encoding proteins that catalyse the transposition process.
- Composite transposons: IS elements flanking resistance genes

- Key in AMR spread (e.g., IS26, Tn3 family)



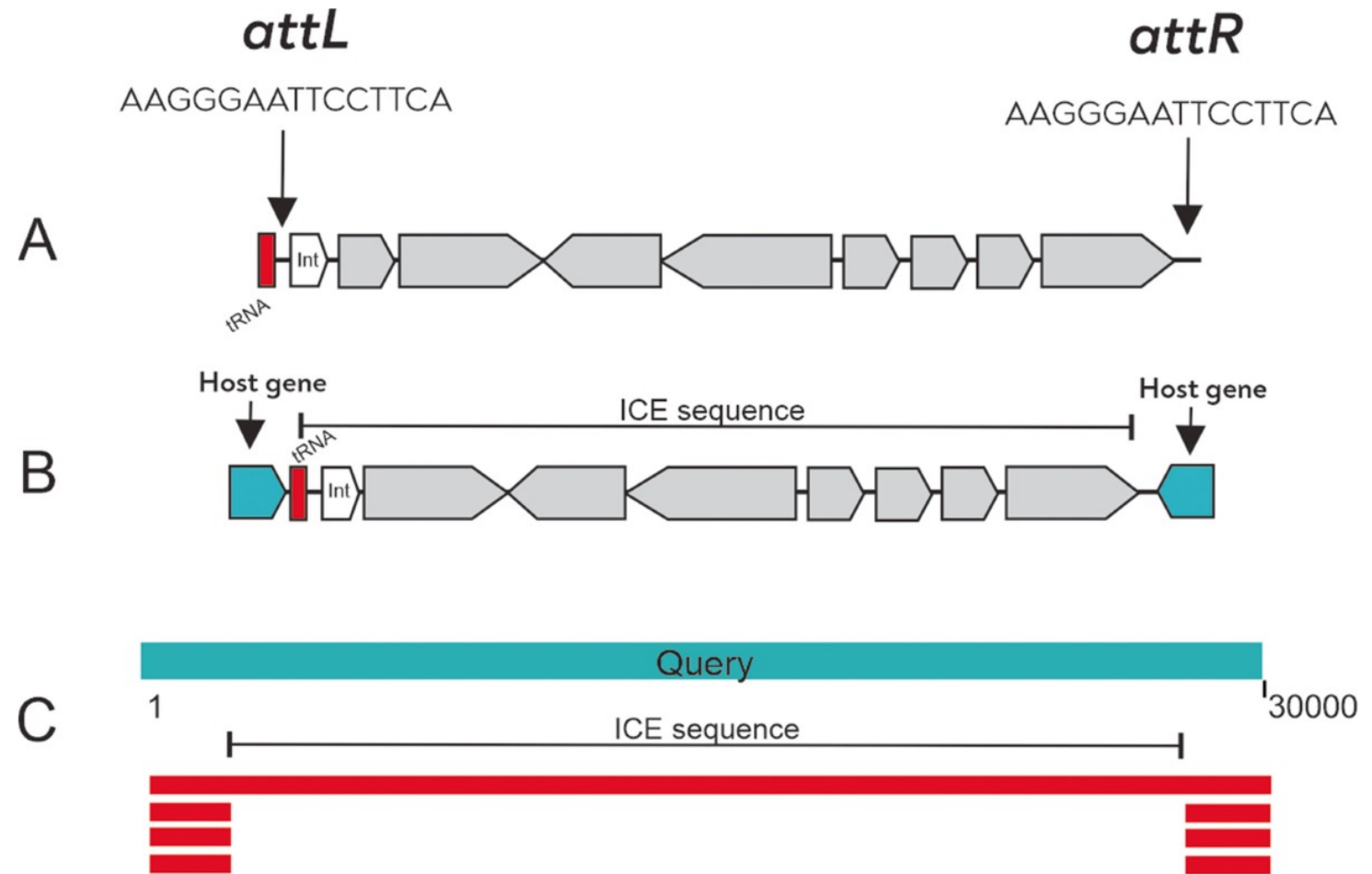
Integrans

- A gene encoding a site-specific recombinase
- A proximal recombination site where cassettes may be inserted, recognisable by the integrase
- One or more incorporated gene cassettes.
- They capture and express genes (usually AMR genes).
- They may be found as part of other MGEs.



Integrative and conjugative elements

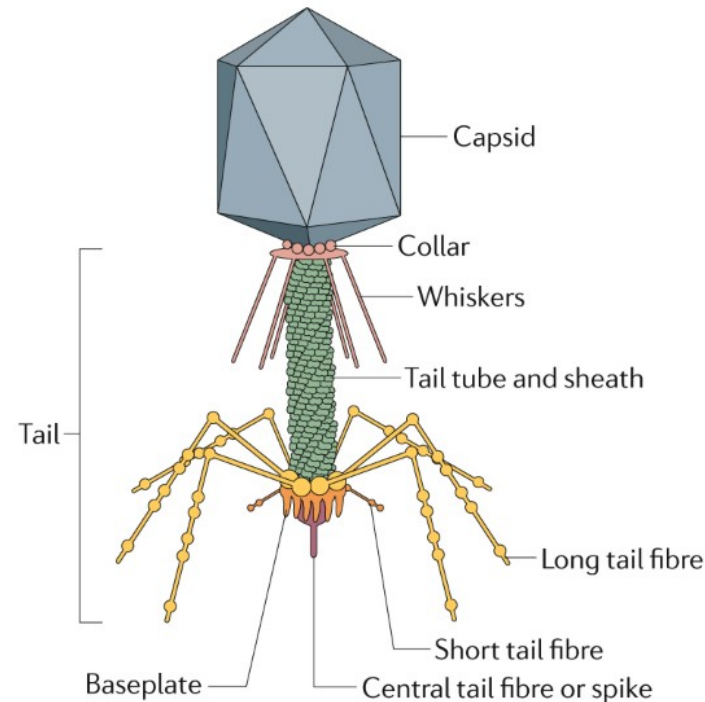
- ICEs are mobile genetic elements that can integrate into the bacterial genome and transfer themselves to other bacteria via conjugation.
- They are located primarily on the chromosome but have the ability to excise themselves.
- ICEs combine the features of bacteriophages and transposons.
- An example is the SXT ICE in *V. cholerae*



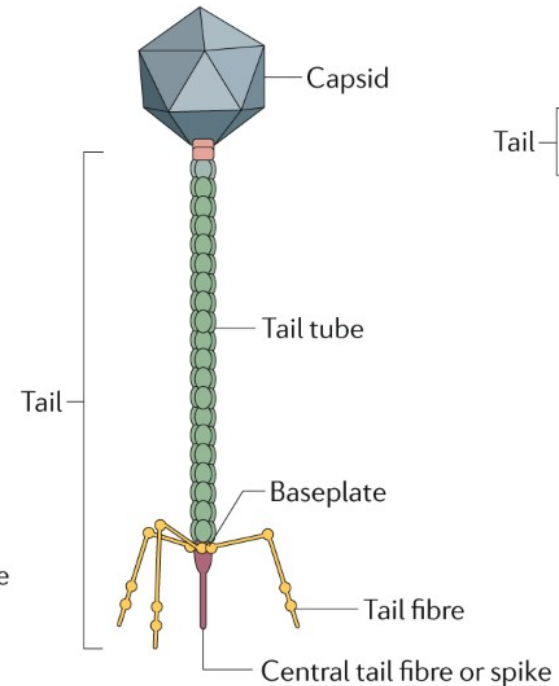
Bacteriophages

- Viruses infecting bacteria
- Two main types: lytic (virulent), lysogenic (temperate)
- Integration as prophages in host genome
- Influence bacterial fitness and virulence
- They can transfer bacterial genes between bacterial cells by transduction.
- An example is the lambda phage, it integrates into the bacterial genome and can carry bacterial genes.

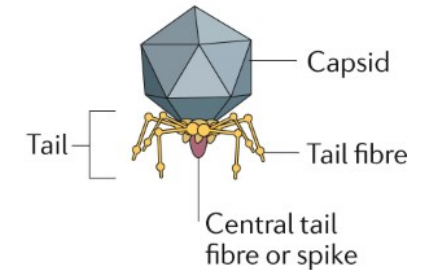
a Myoviridae



b Siphoviridae



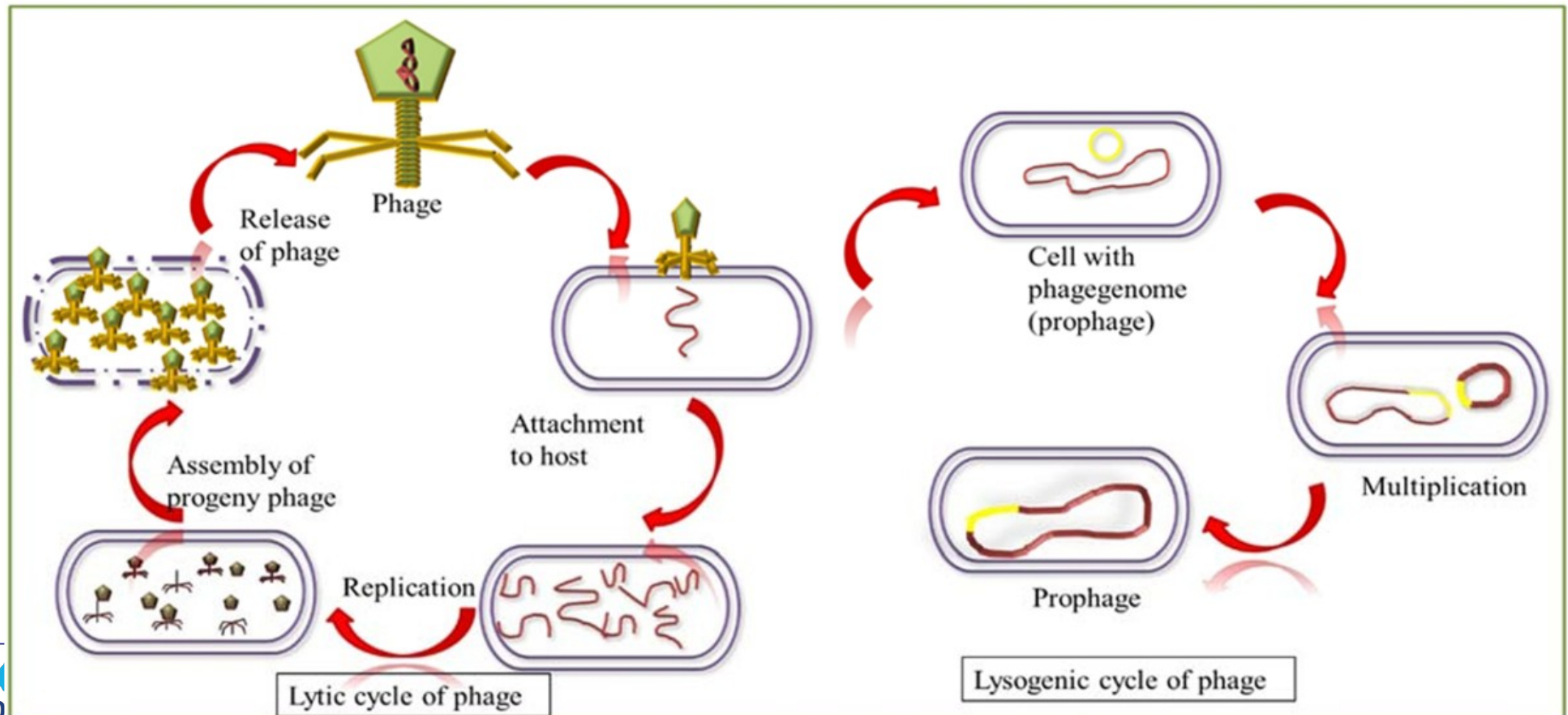
c Podoviridae



Phage Life Cycles

- **Lytic cycle:** Typically involve attaching to a host cell, injecting their genetic material, and then immediately destroying the cell to release new phages

- **Lysogenic cycle:** Integrating their genetic material into the host's genome to replicate along with it



MGE's Relevance to Bacterial Genomics

- MGEs often work together: plasmid–transposon–integron–phage networks
- Promote rapid spread of AMR genes
- Drive genome evolution and adaptation
- MGEs shape bacterial population structure
- Crucial for outbreak tracking and molecular epidemiology
- Targets for genomic surveillance and AMR monitoring
- Basis for novel therapeutics (e.g., phage therapy)

Identification of MGEs

- Identification of MGEs from sequencing data is mainly by use of computational approaches to detect specific features or sequences associated with these mobile genetic elements.
- Methods of identification rely heavily on databases of already identified MGEs or of key MGE features.
- Different tools are used to detect presence, identify and classify these MGEs.

Identification of MGEs - Tools

Tools for detection

- MOB-suite
- **PlasmidFinder**
- IslandViewer
- ICEberg
- **Prokka**
- MetaCHIP
- TranspoosnPSI
- CONJscan
- **PHAge Search Tool (PHAST)**
- PRACTICAL
- MobileElementFinder
- **ICEfinder**
- Plasmid Database (PLSDB)
- TnCentral
- OriTfinder
- **ISFinder**
- PLACNETw
- **IntegronFinder, PHASTER, ISEScan**